Niloofar Mireshghallah

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RESEARCH INTERESTS

- Privacy, contextual integrity & information-flow norms
- Memorization, membership inference & data integrity
- AI for science: chemistry & drug discovery
- AI, mental health & user wellbeing

POSITIONS

Assistant Professor
Carnegie Mellon University *Aug 2025–Present*
Joint appointment: Language Technologies Institute (LTI) & Engineering and Public Policy (EPP); courtesy appointment in Software and Societal Systems (S3D); core member of CyLab.

Founding Member of Technical Staff
humans& ai *Nov 2025–Nov 2026*
User Model Development and Evaluation, RL Environment Development

Research Scientist
Meta AI (*FAIR Alignment: Privacy & Security; collab. with FAIR Chemistry*) *May 2025–Nov 2025*
Privacy-preserving AI systems and LLM safety.

Postdoctoral Scholar
University of Washington *May 2023–May 2025*
Advisors: Yejin Choi, Yulia Tsvetkov.

Research Intern / Part-time Researcher
Microsoft (*Semantic Machines; MSR Algorithms; MSR Language & Intelligent Assistance*) *Jun 2020–Jul 2023*
Mentors: Richard Shin, Yu Su, Tatsunori Hashimoto, Jason Eisner, Sergey Yekhanin, Arturs Backurs, Dimitrios Dimitriadis, Robert Sim, Huseyin Inan.

Research Intern
Western Digital Co. Research and Development *Summer 2019*

EDUCATION

Ph.D. & M.S. in Computer Science
University of California San Diego, USA *Sep 2018–Apr 2023*
Advisor: Taylor Berg-Kirkpatrick. M.S. awarded Jun 2020. CGPA 3.90/4.00.

B.Sc. in Computer Engineering
Sharif University of Technology, Iran *Sep 2014–Jun 2018*

CGPA 18.12/20.00.

AWARDS & HONORS

Delta Institute Scientific Fellow *2026*
Greenwall Foundation Faculty Scholars Program in Bioethics, Finalist *2026*
Momental Foundation Mistletoe Research Fellowship (MRF), Finalist *2023*
Rising Star in Adversarial Machine Learning (AdvML), Award Winner *2022*
Rising Stars in EECS *2022*
UCSD CSE Excellence in Leadership and Service, Award Winner *2022*
FAccT Doctoral Consortium *2022*
Qualcomm Innovation Fellowship, Finalist *2021*
NCWIT (National Center for Women & IT) Collegiate Award, Winner *2020*
National University Entrance Exam in Math (Ranked 249th of 223,000) *2014*
National University Entrance Exam in Foreign Languages (Ranked 57th of 119,000) *2014*
National Organization for Exceptional Talents (NODET), Admitted (∼2% Acceptance Rate) *2008*

GRANTS & FUNDING

Research Grants & Gifts

DOE Genesis Mission — collaboration with Brookhaven National Laboratory *2026*
Longview Philanthropy — AI hardware security & verification of international AI agreements *2026*
Fujitsu — research grant *2026*
OpenAI — Mental Health Research Grant *2026*
Anthropic — AI for Science Program *2026*
Foresight Institute — AI for Science & Safety *2026*

Compute & Platform Grants

Google TPU Research Cloud (TRC) *2026*
Google Gemini compute grant *2026*
Thinking Machines (Tinker) Academic Research Compute Grant *2025*
Modal Academic Research Compute Grant *2025*
Additional compute support: Lambda, River, Prime Intellect Academic Research Compute Grants *2025–2026*

PUBLICATIONS

* denotes equal contribution. For the full list, see Google Scholar.

Preprints & Under Review

1. A. Priyanshu, S. Vijay, B. Jabarian, and **N. Mireshghallah**, "Et Tu, Brute? Economic Misalignment in Personal AI Agents," *arXiv:2609.24927*, 2026.

2. S. Majidi, **N. Mireshghallah**, and K. Taram, "Leaky Language Models: Stealing Architecture and Inference Optimizations via Per-Token Timing," *arXiv:2607.20723*, 2026.

3. A. Hughes, A. Goldberg, P. Jha, A. Perer, N. Aletras, and **N. Mireshghallah**, "Boundary-Targeted Membership Inference Attacks on Safety Classifiers," *arXiv:2605.22373*, 2026.

4. K. Han, R. Zhang, K. Wei, H. Mahdavi, **N. Mireshghallah**, and A. Barati Farimani, "SMDD-Bench: Can LLMs Solve Real-World Small Molecule Drug Design Tasks?," *arXiv:2605.21740*, 2026. Poster at the GenBio Workshop, ICML 2026.

5. K. Monteiro, M. Park, A. Ioffrida, A. Sanna, H.-P. Lee, **N. Mireshghallah**, Y. Wang, and S. Das, "When Are LLM Inferences Acceptable? User Reactions and Control Preferences for Inferred Personal Information," *arXiv:2605.10013*, 2026.

6. X. Liu, **N. Mireshghallah**, J. C. Ginsburg, and T. Chakrabarty, "Alignment Whack-a-Mole: Finetuning Activates Verbatim Recall of Copyrighted Books in Large Language Models," *arXiv:2603.20957*, 2026. **Oral** at the MemFM Workshop, ICML 2026.

7. J. X. Morris, **N. Mireshghallah**, M. Ibrahim, and S. Mahloujifar, "Learning to Reason in 13 Parameters," *arXiv:2602.04118*, 2026.

8. J. Borkar, K. Chadha, **N. Mireshghallah**, Y. Zhang, I.-E. Veliche, A. Mitra, D. A. Smith, Z. Xu, and D. Garcia-Olano, "Memorization Dynamics in Knowledge Distillation for Language Models," *arXiv:2601.15394*, 2026.

9. R. Schaeffer, J. Kazdan, B. Abbasi, K. Z. Liu, B. Miranda, A. Ahmed, F. Barez, A. Puri, S. Biderman, **N. Mireshghallah**, and S. Koyejo, "Quantifying the Effect of Test Set Contamination on Generative Evaluations," *arXiv:2601.04301*, 2026. Poster at the FoGen Workshop, ICML 2026.

10. P. Mahdavinia, H. Mahdavi, **N. Mireshghallah**, and M. Mahdavi, "Harnessing Optimization Dynamics for Curvature-Informed Model Merging," *arXiv:2509.11167*, 2025.

11. J. Roh, Z. Novack, Y. Peng, **N. Mireshghallah**, T. Berg-Kirkpatrick, and A. Houmansadr, "Bob's Confetti: Phonetic Memorization Attacks in Music and Video Generation," *arXiv:2507.17937*, 2025.

12. A. Naseh and **N. Mireshghallah**, "Synthetic Data Can Mislead Evaluations: Membership Inference as Machine Text Detection," *arXiv:2501.11786*, 2025.

13. M. H. Garcia, A. Manoel, D. M. Diaz, **N. Mireshghallah**, R. Sim, and D. Dimitriadis, "FLUTE: A Scalable, Extensible Framework for High-Performance Federated Learning Simulations," *arXiv:2203.13789*, 2022.

14. **N. Mireshghallah**, M. Taram, P. Vepakomma, A. Singh, R. Raskar, and H. Esmaeilzadeh, "Privacy in Deep Learning: A Survey," *arXiv:2004.12254*, 2020.

Conference Papers

15. R. Zhang, M. Kaniselvan, and **N. Mireshghallah**, "Reinforcement Learning Improves Traversal of Parametric Knowledge in LLMs," *EMNLP 2026* (**Oral**).

16. H. Kim*, **N. Mireshghallah***, M. Duan, R. Xin, S. S. Li, J. Jung, D. Acuna, Q. Pang, H. Xiao, G. E. Suh, S. Oh, Y. Tsvetkov, P. W. Koh, and Y. Choi, "Privasis: Synthesizing the Largest 'Public' Private Dataset from Scratch," *ICML 2026*.

17. S. Biderman, M. A. Khan, **N. Mireshghallah**, C. Arnett, F. Barez, and N. Saphra, "Position: Don't Just 'Fix it in Post': A Science of AI Must Study Learning Dynamics," *ICML 2026* (**Oral**).

18. **N. Mireshghallah**, N. Mangaokar, N. Kokhlikyan, A. Zharmagambetov, M. Zaheer, S. Mahloujifar, and K. Chaudhuri, "CIMemories: A Compositional Benchmark for Contextual Integrity of Persistent Memory in LLMs," *ICLR 2026*.

19. J. Zhou, **N. Mireshghallah**, and T. Li, "Operationalizing Data Minimization for Privacy-Preserving LLM Prompting," *ICLR 2026*.

20. T. Sorensen, B. Newman, J. Moore, C. Y. Park, J. Fisher, **N. Mireshghallah**, L. Jiang, and Y. Choi, "Spectrum Tuning: Post-Training for Distributional Coverage and In-Context Steerability," *ICLR 2026*.

21. D. Pham, P. Kairouz, **N. Mireshghallah**, E. Bagdasarian, C. M. Pham, and A. Houmansadr, "Can Large Language Models Really Recognize Your Name?," *FAccT 2026*.

22. R. Xin*, **N. Mireshghallah***, S. S. Li, M. Duan, H. Kim, Y. Choi, Y. Tsvetkov, S. Oh, and P. W. Koh, "A False Sense of Privacy: Evaluating Textual Data Sanitization Beyond Surface-Level Privacy Leakage," *SaTML 2026*.

23. A. F. Cooper, ..., **N. Mireshghallah**, ..., and K. Lee, "Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy and Research," *NeurIPS 2025* (**Oral**).

24. J. Hayes, I. Shumailov, C. A. Choquette-Choo, M. Jagielski, G. Kaissis, M. Nasr, M. S. M. S. Annamalai, **N. Mireshghallah**, I. Shilov, M. Meeus, Y.-A. de Montjoye, K. Lee, F. Boenisch, A. Dziedzic, and A. F. Cooper, "Exploring the Limits of Strong Membership Inference Attacks on Large Language Models," *NeurIPS 2025*.

25. T. Chen, F. Brahman, J. Liu, **N. Mireshghallah**, W. Shi, P. W. Koh, L. Zettlemoyer, and H. Hajishirzi, "ParaPO: Aligning Language Models to Reduce Verbatim Reproduction of Pre-training Data," *COLM 2025*.

26. S. Hallinan, J. Jung, M. Sclar, X. Lu, A. Ravichander, S. Ramnath, Y. Choi, S. P. Karimireddy, **N. Mireshghallah**, and X. Ren, "The Surprising Effectiveness of Membership Inference with Simple N-Gram Coverage," *COLM 2025*.

27. X. Zhou, H. Kim, F. Brahman, L. Jiang, H. Zhu, X. Lu, F. Xu, B. Y. Lin, Y. Choi, **N. Mireshghallah**, R. Le Bras, and M. Sap, "HAICOSYSTEM: An Ecosystem for Sandboxing Safety Risks in Interactive AI Agents," *COLM 2025*.

28. J. Borkar, M. Jagielski, K. Lee, **N. Mireshghallah**, D. A. Smith, and C. A. Choquette-Choo, "Privacy Ripple Effects from Adding or Removing Personal Information in Language Model Training," *Findings of ACL 2025*. Cited in US Congressional testimony on AI chatbot data privacy.

29. X. Lu, M. Sclar, S. Hallinan, **N. Mireshghallah**, J. Liu, S. Han, A. Ettinger, L. Jiang, K. R. Chandu, N. Dziri, and Y. Choi, "AI as Humanity's Salieri: Quantifying Linguistic Creativity of Language Models via Systematic Attribution of Machine Text against Web Text," *ICLR 2025* (**Oral**).

30. I. C. Ngong, J. P. Near, and **N. Mireshghallah**, "Differentially Private Learning Needs Better Model Initialization and Self-Distillation," *NAACL 2025* (**Oral**).

31. A. Ravichander, J. Fisher, T. Sorensen, X. Lu, M. Antoniak, B. Y. Lin, **N. Mireshghallah**, C. Bhagavatula, and Y. Choi, "Information-Guided Identification of Training Data Imprint in (Proprietary) Large Language Models," *NAACL 2025* (**Oral; Honorable Mention candidate**).

32. A. M. Kassem*, O. Mahmoud*, **N. Mireshghallah***, H. Kim, Y. Tsvetkov, Y. Choi, S. Saad, and S. Rana, "Alpaca against Vicuna: Using LLMs to Uncover Memorization of LLMs," *NAACL 2025*.

33. L. Jiang, K. Rao, S. Han, A. Ettinger, F. Brahman, S. Kumar, **N. Mireshghallah**, X. Lu, M. Sap, Y. Choi, and N. Dziri, "WildTeaming at Scale: From In-the-Wild Jailbreaks to (Adversarially) Safer Language Models," *NeurIPS 2024*.

34. T. Chen, A. Asai*, **N. Mireshghallah***, S. Min, J. Grimmelmann, Y. Choi, H. Hajishirzi, L. Zettlemoyer, and P. W. Koh, "CopyBench: Measuring Literal and Non-Literal Reproduction of Copyright-Protected Text in Language Model Generation," *EMNLP 2024*.

35. M. Duan, A. Suri, **N. Mireshghallah**, S. Min, W. Shi, L. Zettlemoyer, Y. Tsvetkov, Y. Choi, D. Evans, and H. Hajishirzi, "Do Membership Inference Attacks Work on Large Language Models?," *COLM 2024*.

36. **N. Mireshghallah***, M. Antoniak*, Y. More*, Y. Choi, and G. Farnadi, "Trust No Bot: Discovering Personal Disclosures in Human–LLM Conversations in the Wild," *COLM 2024*.

37. T. Sorensen, J. Moore, J. Fisher, M. Gordon, **N. Mireshghallah**, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, T. Althoff, and Y. Choi, "Position: A Roadmap to Pluralistic Alignment," *ICML 2024*.

38. **N. Mireshghallah***, H. Kim*, X. Zhou, Y. Tsvetkov, M. Sap, R. Shokri, and Y. Choi, "Can LLMs Keep a Secret? Testing Privacy Implications of Language Models via Contextual Integrity Theory," *ICLR 2024* (**Spotlight**). Used by Anthropic and Meta as a pre-deployment privacy evaluation.

39. X. Tang, R. Shin, H. A. Inan, A. Manoel, **N. Mireshghallah**, Z. Lin, S. Gopi, J. Kulkarni, and R. Sim, "Privacy-Preserving In-Context Learning with Differentially Private Few-Shot Generation," *ICLR 2024*.

40. M. Zhang, T. He, T. Wang, L. Mi, **N. Mireshghallah**, B. Chen, H. Wang, and Y. Tsvetkov, "LatticeGen: Hiding Generated Text in a Lattice for Privacy-Aware Large Language Model Generation on Cloud," *Findings of NAACL 2024*.

41. **N. Mireshghallah**, J. Mattern, S. Gao, R. Shokri, and T. Berg-Kirkpatrick, "Smaller Language Models are Better Zero-shot Machine-Generated Text Detectors," *EACL 2024*.

42. **N. Mireshghallah**, Y. Su, T. Hashimoto, J. Eisner, and R. Shin, "Privacy-Preserving Domain Adaptation of Semantic Parsers," *ACL 2023*.

43. J. Mattern, **N. Mireshghallah**, Z. Jin, B. Schölkopf, M. Sachan, and T. Berg-Kirkpatrick, "Membership Inference Attacks against Language Models via Neighbourhood Comparison," *Findings of ACL 2023*.

44. **N. Mireshghallah***, N. Vogler*, J. He, O. Florez, A. El-Kishky, and T. Berg-Kirkpatrick, "Simple Temporal Adaptation to Changing Label Sets: Hashtag Prediction via Dense KNN," *EMNLP 2023*.

45. J. Forristal, **N. Mireshghallah**, G. Durrett, and T. Berg-Kirkpatrick, "A Block Metropolis-Hastings Sampler for Controllable Energy-Based Text Generation," *CoNLL 2023*.

46. **N. Mireshghallah**, A. Backurs, H. A. Inan, L. Wutschitz, and J. Kulkarni, "Differentially Private Model Compression," *NeurIPS 2022*.

47. **N. Mireshghallah**, A. Uniyal, T. Wang, D. Evans, and T. Berg-Kirkpatrick, "An Empirical Analysis of Memorization in Fine-tuned Autoregressive Language Models," *EMNLP 2022* (**Oral**).

48. **N. Mireshghallah**, K. Goyal, A. Uniyal, T. Berg-Kirkpatrick, and R. Shokri, "Quantifying Privacy Risks of Masked Language Models Using Membership Inference Attacks," *EMNLP 2022*.

49. H. Brown, K. Lee, **N. Mireshghallah**, R. Shokri, and F. Tramèr, "What Does it Mean for a Language Model to Preserve Privacy?," *FAccT 2022*.

50. **N. Mireshghallah**, V. Shrivastava, M. Shokouhi, T. Berg-Kirkpatrick, R. Sim, and D. Dimitriadis, "UserIdentifier: Implicit User Representations for Simple and Effective Personalized Sentiment Analysis," *NAACL 2022*.

51. **N. Mireshghallah**, K. Goyal, and T. Berg-Kirkpatrick, "Mix and Match: Learning-free Controllable Text Generation using Energy Language Models," *ACL 2022*.

52. **N. Mireshghallah** and T. Berg-Kirkpatrick, "Style Pooling: Automatic Text Style Obfuscation for Improved Classification Fairness," *EMNLP 2021* (**Oral**).

53. **N. Mireshghallah**, H. A. Inan, M. Hasegawa, V. Rühle, T. Berg-Kirkpatrick, and R. Sim, "Privacy Regularization: Joint Privacy–Utility Optimization in Language Models," *NAACL 2021*.

54. **N. Mireshghallah**, M. Taram, A. Jalali, A. T. Elthakeb, D. Tullsen, and H. Esmaeilzadeh, "Not All Features Are Equal: Discovering Essential Features for Preserving Prediction Privacy," *WWW 2021*.

55. T. Koker, **N. Mireshghallah**, T. Titcombe, and G. Kaissis, "U-Noise: Learnable Noise Masks for Interpretable Image Segmentation," *ICIP 2021*.

56. A. T. Elthakeb, P. Pilligundla, **N. Mireshghallah**, A. Cloninger, and H. Esmaeilzadeh, "Divide and Conquer: Leveraging Intermediate Feature Representations for Quantized Training of Neural Networks," *ICML 2020*.

57. **N. Mireshghallah**, M. Taram, P. Ramrakhyani, A. Jalali, D. Tullsen, and H. Esmaeilzadeh, "Shredder: Learning Noise Distributions to Protect Inference Privacy," *ASPLOS 2020*.

Journal Papers

58. A. T. Elthakeb, P. Pilligundla, **N. Mireshghallah**, A. Yazdanbakhsh, and H. Esmaeilzadeh, "ReLeQ: A Reinforcement Learning Approach for Automatic Deep Quantization of Neural Networks," *IEEE Micro*, 2020.

59. **N. Mireshghallah**, M. Bakhshalipour, M. Sadrosadati, and H. Sarbazi-Azad, "Energy-Efficient Permanent Fault Tolerance in Hard Real-Time Systems," *IEEE Transactions on Computers*, 2019.

Workshop & Tutorial Papers

60. R. Zhang and **N. Mireshghallah**, "Reinforcement Learning on Benign Facts Amplifies Leakage of Memorized Private Data," *TAE Workshop at NeurIPS 2026*. arXiv:2608.21727.

61. E. Farhana, M. Zainab, Q. Wang, **N. Mireshghallah**, R. van der Meulen, and M. J. van Duijn, "Theory of Mind and Application in Educational Context," *BEA Workshop at ACL 2026* (tutorial).

62. H. Mahdavi, P. Mahdavinia, S. Malek, P. Mohammadipour, A. Hashemi, M. Daliri, A. Farhadi, A. Khasahmadi, **N. Mireshghallah**, and V. G. Honavar, "RefGrader: Automated Grading of Mathematical Competition Proofs using Agentic Workflows," *NeurIPS 2025 Workshop on MATH-AI*.

63. Y. Bae, M. Kim, J. Lee, S. Kim, J. Kim, Y. Choi, and **N. Mireshghallah**, "Privacy-Preserving LLM Interaction with Socratic Chain-of-Thought Reasoning and Homomorphically Encrypted Vector Databases," *Workshop on Privacy-Preserving Machine Learning (PPML) at CRYPTO 2025*.

64. N. G. Brigham, C. Gao, T. Kohno, F. Roesner, and **N. Mireshghallah**, "Breaking News: Case Studies of Generative AI's Use in Journalism," *SoLaR Workshop at NeurIPS 2024*.

65. K. Lee, A. F. Cooper, C. A. Choquette-Choo, K. Liu, M. Jagielski, **N. Mireshghallah**, et al., "Machine Unlearning Doesn't Do What You Think," *GenLaw Workshop at ICML 2024* (Extended Abstract).

66. R. Naidu, A. Priyanshu, A. Kumar, S. Kotti, H. Wang, and **N. Mireshghallah**, "When Differential Privacy Meets Interpretability: A Case Study," *Responsible Computer Vision Workshop at CVPR 2021*.

67. P. Basu, T. Singha Roy, R. Naidu, Z. Müftüoglu, S. Singh, and **N. Mireshghallah**, "Benchmarking Differential Privacy and Federated Learning for BERT Models," *Machine Learning for Data Workshop at ICML 2021*.

68. A. Uniyal, R. Naidu, S. Kotti, S. Singh, P. J. Kenfack, **N. Mireshghallah**, and A. Trask, "DP-SGD vs. PATE: Which Has Less Disparate Impact on Model Accuracy?," *Machine Learning for Data Workshop at ICML 2021*.

69. T. Farrand, **N. Mireshghallah**, S. Singh, and A. Trask, "Neither Private Nor Fair: Impact of Data Imbalance on Utility and Fairness in Differential Privacy," *PPMLP Workshop at CCS 2020*.

Patents

70. J. M. Eisner, E. C. Shin, **N. Mireshghallah**, T. B. Hashimoto, and Y. Su, "Privacy-Preserving Generation of Synthesized Training Data," US Patent Application 18/321,460 (2024).

71. **N. Mireshghallah** and H. Esmaeilzadeh, "Methods of Providing Data Privacy for Neural Network-Based Inference," US Patents 11,288,379; 11,487,884; and 12,210,971.

72. **N. Mireshghallah**, H. Esmaeilzadeh, and M. Taram, "Methods and Systems for Learning Noise Distributions for Neural Networks," US Patent 12,008,470 (granted Jun. 2024).

Policy & Technical Write-ups

73. **N. Mireshghallah** and T. Li, "Position: Privacy Is Not Just Memorization!," *arXiv:2510.01645*, 2025. Featured in Help Net Security.

74. A. F. Cooper, K. Lee, J. Grimmelmann, D. Ippolito, C. Callison-Burch, C. A. Choquette-Choo, **N. Mireshghallah**, et al., "Report of the 1st Workshop on Generative AI and Law," *Yale Law & Economics Research Paper, SSRN-4634513*, 2023.

INVITED TALKS

Keynotes & Recent Invited Talks

Copenhagen NLP Symposium
Keynote *Oct. 2026*
What Goes Where? Information Flow in Long-Horizon AI Agents.

RE-Data: Workshop on Responsibly Enabling Data for Foundation Models at COLM 2026
Invited Talk *Oct. 2026*
How Does Post-Training Change the Imprint of Pretraining Data?

2nd Workshop on Human-Centered Privacy and Security for Language Models (HAIPS) at COLM 2026
Panel Moderator *Oct. 2026*

Panel with Helen Nissenbaum, Ruoxi Jia, Maarten Sap, and Nina Taft.

Carnegie Mellon University, School of Computer Science
New Faculty Lightning Talk *Aug. 2026*
Learning the Tails!

ICML 2026 (Seoul)
Invited Debater, Workshop on AI as a Tool for Mathematics, CS & ML *Jul. 2026*
Debate with Ludwig Schmidt: will society keep paying humans to do research?

ICML 2026 (Seoul)
Panelist, WiML Symposium ("Career Transitions in ML") and KAIST@ICML ("Career Advice for Junior Researchers") *Jul. 2026*

Global AI Frontiers Symposium (Seoul)
Panelist *Jul. 2026*

BEA Workshop at ACL 2026
Tutorial Co-instructor *Jul. 2026*
Theory of Mind and Application in Educational Context.

Google
Invited Talk *Jun. 2026*

University of Cagliari (PRA Lab), Machine Learning Security Seminar Series
Invited Talk *Jun. 2026*
What Should an Agent Remember, Reveal, and Redact? The New Data Sinks and Leaks of Agentic AI.

Cornell University
AI Seminar *May 2026*
2026 Is the New 2016, but Make It Privacy: On Federated Memory, Contextual Privacy, and Personalized Agents.

MIT CSAIL
NLP Seminar *Apr. 2026*
2026 Is the New 2016, but Make It Privacy: On Federated Memory, Contextual Privacy, and Personalized Agents.

Simons Institute, UC Berkeley
Keynote, Workshop on Trust in Decentralized Systems *Mar. 2026*
2026 Is the New 2016, but Make It Privacy: On Federated Memory, Contextual Privacy, and Personalized Agents.

FAR AI San Diego Alignment Workshop at NeurIPS
Invited Talk *Dec. 2025*
What Does It Mean for Agentic AI to Preserve Privacy?

Conference on Applied Machine Learning in Information Security (CAMLIS)
Keynote *Oct. 2025*

What Does It Mean for Agentic AI to Preserve Privacy? Mapping the New Data Sinks and Leaks.

Cornell Tech
Digital Life Seminar *Oct. 2025*
Contextual Privacy in LLMs: Benchmarking and Mitigating Inference-Time Risks.

Meta AI / FAIR Alignment Group
Invited Talk *Oct. 2025*
What You Should *Really* Worry About When It Comes to Generative AI and Privacy.

First Workshop on LLM Security (LLMSec) at ACL 2025
Keynote *Aug. 2025*
What Does It Mean for Agentic AI to Preserve Privacy?

First Workshop on LLM Memorization (L2M2) at ACL 2025
Keynote *Aug. 2025*
Emergent Misalignment Through the Lens of Non-verbatim Memorization.

Workshop on Collaborative and Federated Agentic Workflows (CFAgentic) at ICML 2025
Invited Talk *Jul. 2025*

Fifth Workshop on Trustworthy NLP (TrustNLP) at NAACL 2025
Invited Talk *May 2025*
A False Sense of Privacy: Semantic Leakage and Non-literal Copying in LLMs.

ETH Zurich
Guest Lecture *May 2025*
Privacy, Copyright and Data Integrity: The Cascading Implications of Generative AI.

UC Berkeley School of Information
Invited Talk *Feb. 2025*
Privacy, Copyright, and Data Integrity: The Cascading Implications of Generative AI.

Stanford University
NLP Seminar *Jan. 2025*
Privacy, Copyright and Data Integrity: The Cascading Implications of Generative AI.

Academic Job Talks (Jan.–Mar. 2025)
Privacy, Copyright and Data Integrity: The Cascading Implications of Generative AI.
Johns Hopkins University, UC Berkeley, Carnegie Mellon University, UW–Madison, UPenn, Georgia Tech, UCLA, University of Maryland, University of Michigan, NYU, EPFL, ETH Zurich, UT Austin, UVA, UNC Chapel Hill.

2024

NeurIPS 2024 Red Teaming GenAI Workshop
Keynote *Dec. 2024*
A False Sense of Privacy: Semantic Leakage and Non-literal Copying in LLMs.

NeurIPS 2024 PrivacyML Tutorial

Panelist *Dec. 2024*

Johns Hopkins University
CS Department Seminar *Dec. 2024*
Privacy, Copyright and Data Integrity: The Cascading Implications of Generative AI.

Future of Privacy Forum
Panelist, Technologist Roundtable for Policymakers *Nov. 2024*
Key Issues in Privacy and AI.

National Academies (NASEM)
Forum on Cyber Resilience *Aug. 2024*
Oversharing with LLMs is Underrated.

University seminars & guest lectures (2024): UMass Amherst NLP Seminar; Northeastern Khoury Security Seminar; UC Santa Barbara & UCLA NLP Seminars; Georgia Tech School of Interactive Computing; University of Utah; UW Allen School Colloquium; UT Austin; CMU (LTI 11-830 & invited talk).

Industry & research-org talks (2024): Google Brain; Meta AI Research; LinkedIn Research; SRI International C3E Workshop; Simons Collaboration TOC4Fairness Seminar; Generative AI & Law Workshop (Washington, DC); ML Collective DLCT.

Earlier Talks (selected)

CISPA Helmholtz Center for Information Security (2023); Max Planck Institute for Software Systems (2023); NDSS EthiCS Workshop — Keynote (2023); Google Federated Learning Seminar (2023); Mila / McGill University (2023); EACL 2023 Tutorial Co-instructor; UW eScience Institute (2023); Cohere for AI (2023).

KDD AdvML Workshop (2022); UT Austin (2022); Microsoft Research Cambridge (2022); Johns Hopkins University Guest Lecture (2022); PriSec ML Interest Group (2022).

Alan Turing Institute (2021); National University of Singapore (2021); Big Science for LLMs Panel (2021); Split Learning Workshop (2021); University of Maine Guest Lecture (2021).

UMass Amherst ML and Friends Lunch (2020); OpenMined Privacy Conference (2020); Microsoft Research AI Breakthroughs Workshop (2020); NCWIT Collegiate Award Winner Talk (2020).

STUDENT MENTORING & ADVISING

Current — Looni Lab, Carnegie Mellon University

Bill Watson, Ph.D. student (co-advised with Sauvik Das)
Xiaoyu (Nicholas) Wu, Ph.D. student (co-advised with Steven Wu)
Renfei Zhang, Ph.D. student
Kevin Han, Ph.D. student (co-advised with Amir Barati Farimani)
Bhuvan Chandra Koduru, Master's student
Visiting researchers & student collaborators: Prince Jha, Anthony Hughes

Past Mentees (2020–2025)

Affiliations as of the mentorship period; arrows show subsequent placement.
Skyler Hallinan, BS/MS student at University of Washington → PhD student at USC
Tong Chen, PhD student at University of Washington
Sara Kodeiri, MS student at University of Waterloo
Sabrina Mokhtari, MS student at University of Waterloo
Rui Xin, PhD student at University of Washington
Yash More, MS student at McGill University
Aly Kassem, MS student at University of Windsor → Research Engineer at Dynamo AI

Michael Duan, BS student at University of Washington → Software Engineer at Uber
Grace Brigham, BS student at University of Washington
Chongjiu Gao, BS/MS student at University of Washington
Supriti Vijay, BS at Manipal Institute → MS student at Carnegie Mellon University
Justus Mattern, BS student at RWTH Aachen → Y Combinator
Ivoline Ngong, PhD student at University of Vermont
Jarad Forristal, BS at University of Texas Austin → PhD student at UC San Diego
Archit Uniyal, BE at UIET Panjab University → PhD student at University of Virginia
Aman Priyanshu, BS at Manipal Institute → MS student at Carnegie Mellon University
Rakshit Naidu, BS at Manipal Institute → PhD student at Georgia Tech
Priyam Basu, BS at Manipal Institute → MS student at University of Washington
Zümrüt Müftüoglu, PhD student at Yıldız Teknik Üniversitesi
Teddy Koker, Research Engineer at Lightning AI → PhD student at MIT EECS
Tom Farrand, Deep Learning Engineer at IBM
Jaydeep Borkar, BE at Pune University → PhD student at Northeastern University

TEACHING

CSE Department, UC San Diego (Teaching Assistant)
CSE 151A (Undergraduate ML); CSE 251A (Graduate ML); CSE 276C (Graduate Mathematics for Robotics); CSE 141 (Undergraduate Computer Architecture); CSE 240D (Graduate Accelerator Design for Deep Learning) *2019–2021* **CE**
Department, Sharif University of Technology (Teaching Assistant)
Digital Electronics; Computer Architecture; Signals and Systems; Probability and Statistics; Numerical Methods 2016–2018

ORGANIZED EVENTS & WORKSHOPS

Co-organizer, 2nd Workshop on Human-Centered Privacy and Security for Language Models (HAIPS) at COLM *2026*
Co-organizer, 2nd Workshop on Memorization and Trustworthy Foundation Models (MemFM) at ICML *2026*
Co-organizer, Workshop on Foundations of Deep Generative Models (FoGen) at ICML *2026*
Advisory Committee, 2nd Workshop on Technical AI Governance Research (TAIGR) at ICML *2026*
Co-organizer, Memorization and Trustworthy Foundation Models (MemFM) Workshop at ICML *2025*
Privacy Session Chair, SAGAI Workshop at IEEE S&P *2025*
Co-organizer, Generative AI and Law (GenLaw) Workshop at ICML *2024*
Co-organizer, Privacy Regulation and Protection in Machine Learning Workshop at ICLR *2024*
Co-organizer, Private NLP Workshop at ACL *2024*
Co-organizer, Privacy-Preserving AI (PPAI) Workshop at AAAI *2024*
Co-organizer, Generative AI and Law (GenLaw) Workshop at ICML *2023*
Co-organizer, Widening NLP (WiNLP) Workshop at EMNLP *2022, 2023*
Co-organizer, Private NLP Tutorial at EACL *2023*
Co-organizer, Ethics in NLP Birds of a Feather Session at EMNLP *2022*
Privacy & Fairness Roundtable Lead, AFCP Workshop at NeurIPS *2022*
Co-organizer, Broadening Collaborations in ML Workshop at NeurIPS *2022*
Co-organizer, Private NLP Workshop at NAACL *2022*
Co-organizer, Federated Learning for NLP Workshop at ACL *2022*
Co-organizer, Widening NLP (WiNLP) Workshop at EMNLP *2021*
Co-organizer, Privacy-Preserving Machine Learning (PPML) Workshop at MICCAI *2021*
Co-organizer, Distributed Private Machine Learning (DPML) Workshop at ICLR *2021*
Co-organizer, SoCal Joint ML and NLP Symposium *2021*
Break-out Co-leader, Women in Machine Learning (WiML) Un-workshop at ICML *2020, 2021*

PROFESSIONAL SERVICE

Area Chair, NeurIPS *2026*
Area Chair, COLM *2025, 2026*
Senior Program Committee, AAAI Alignment Track *2027*
Publicity Chair, ACL *2027*
Area Chair, NAACL *2025*
Area Chair, EMNLP *2024*
PC Member, ACM CCS *2024, 2025*
D&I Co-chair, NAACL *2022, 2025*
Co-chair, Widening NLP (WiNLP) *2022–2024*
Reviewer, ICLR (Outstanding Reviewer Award, 2021) *2021–2025*
Reviewer, NeurIPS; ICML *2020–2024*
Reviewer, AISTATS *2025*
Reviewer, ICLR & ICML Workshop Proposals *2025*
Reviewer, COLM; EACL; CHI *2024*
Reviewer, AAAI; FAccT *2023, 2024*
Reviewer, TACL *2020–Present*
Reviewer, TMLR *2022–Present*
Reviewer, IEEE S&P Magazine *2021–Present*
Reviewer, NeurIPS Workshop Proposals; INLG *2023*
Ethics PC Member, EMNLP; Reviewer, NAACL SRW; PC Member, AFCP & TSRML Workshops at NeurIPS *2022*
Shadow PC Member, IEEE S&P; Artifact Evaluation PC, USENIX Security; Reviewer, CCS Posters *2021*
Artifact Evaluation PC, ASPLOS; PC Member, LatinX in AI & WHI Workshops at ICML, MLArchSys at ISCA; Security & Privacy Committee Member and Session Chair, Grace Hopper Celebration *2020*
Reviewer, ACM TACO and IEEE TC Journals *2020–Present*

DIVERSITY & INCLUSION

Mentor, Women in Machine Learning (WiML) Workshop at NeurIPS *2020, 2024, 2025*
Panelist, CMU SCS Panel: Navigating the Academic Job Market *2025*
Mentor, ACL Mentorship: How to Broadcast Your Research to a Wider Audience? *2025*
Mentor, Graduate Women in Computing (GradWIC) at UCSD *2020–2023*
Mentor, UC San Diego Women Organization for Research Mentoring (WORM) in STEM *2019–2023*
Course Instructor, OpenMined Privacy Course *2020*
Mentor, USENIX Security Undergraduate Mentorship Program *2020*
Mentor, ICLR *2021*
Volunteer & Invited Speaker, WiML Workshop and WiMLDS Meetup at NeurIPS *2019*
Mentor, UCSD CSE Early Research Scholars Program (CSE-ERSP) *2018*

FEATURED PRESS & MEDIA

BBC Persian, *60 Minutes* news program — live segment on AI risks and society *Sep. 2026*
MIT Technology Review — "We still don't know how people are really using AI" (coverage of the AI Observatory) *Aug. 2026*
CMU News — "Carnegie Mellon Researchers Lead Three DOE Genesis Mission Awards to Advance the Future of AI-Enabled Scientific Discovery" *Jul. 2026*
The Information Bottleneck podcast — on the future of generative AI *Jan. 2026*
Science News Explores — "5 things to remember when talking to a chatbot" *Nov. 2025*
Help Net Security — "Most AI privacy research looks the wrong way" *Oct. 2025*
Washington Post — on AI hype, evaluation metrics, and how people judge AI capabilities *Aug. 2025*
Jay Shah Podcast — "Differential Privacy, Creativity & Future of AI Research in the LLM Era" *Feb. 2025*

Science — “AI writing is improving, but it still can’t match human creativity” *Dec. 2024*
The Thesis Review podcast (with Sean Welleck) — on auditing and mitigating safety risks in LLMs *2024*
Washington Post — on Google AI overview errors, chatbot usage patterns, and opting out of AI training *2024*
WIRED — “How to Stop Your Data From Being Used to Train AI” *2024*
UW News — on advances in math and reasoning in new ChatGPT versions *2024*